Abbreviations
~M_5 = sign(rpeQuote(mRPE,Good,hbind(pk(sk1_1),hpol(pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))))),akRPE_1)
~M_6 = pk(sk1_1)
~M_7 = hpol(pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))
~M_8 = sign(rpeQuote(mRPE,Good,hbind(pk(sk2_2),hpol(pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))))),akRPE_1)
~M_9 = pk(sk2_2)
~M_10 = hpol(pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))
~X_1 = (P2,~M_8,~M_9,hpol(pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))) = (P2,sign(rpeQuote(mRPE,Good,hbind(pk(sk2_2),hpol(pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))),akRPE_1),pk(sk2_2),hpol(pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))))
~M_11 = sign(policyMsg(pk(sk2_2),pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))),sk1_1)
~X_2 = (P1,~M_5,~M_6,hpol(pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))) = (P1,sign(rpeQuote(mRPE,Good,hbind(pk(sk1_1),hpol(pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))),akRPE_1),pk(sk1_1),hpol(pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))))
~M_12 = sign(policyMsg(pk(sk1_1),pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))),sk2_2)
~M_13 = sign(confirmMsg(pk(sk2_2),hstar(mkstar(OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))),sk1_1)
~M_14 = sign(confirmMsg(pk(sk1_1),hstar(mkstar(OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))),sk2_2)

A trace has been found.

Honest Process

Attacker

{1}new akRPE_1
{2}new akTEE_1
{3}new akADV_1

(~M,~M_1,~M_2,~M_3) = (pk(akRPE_1),pk(akTEE_1), pk(akADV_1),akADV_1)

!

Beginning of process rpe1

{9}new sk1_1

{11}event PolicyDeclared(P1,pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

Beginning of process rpe2

Beginning of process instance

{180}new skE_2

~M_4 = pk(skE_2)

(P1,~M_5,~M_6,~M_7)

PermissiveRpeTCB

{95}new sk2_2

{97}event PolicyDeclared(P2,pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

(P2,~M_8,~M_9,~M_10)

~X_1

{20}event EvidenceChecked(P1,P2,pk(sk2_2),hpol(pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{21}event QuotesComplete(P1)

{22}event PolicySent(P1,pk(sk2_2),pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exacts1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exacts2),exact(prodA),V1,OnlyGood,Assign(P2)))

(P1,P2,~M_11)

~X_2

{106}event EvidenceChecked(P2,P1,pk(sk1_1),hpol(pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exacts1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{107}event QuotesComplete(P2)

{108}event PolicySent(P2,pk(sk1_1),pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

(P2,P1,~M_12)

(P2,P1,~M_12)

{26}event PolicyAuthenticated(P1,P2,pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{29}event PolicyAccepted(P1,P2,pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{30}event PolicyAcceptedKeys(P1,P2,pk(sk1_1),pk(sk2_2),pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{31}event PoliciesComplete(P1)

{34}event FullJoin(P1,mkstar(OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{35}event ConfirmationSent(P1,P2,hstar(mkstar(OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))))

(P1,P2,~M_13)

(P1,P2,~M_11)

{112}event PolicyAuthenticated(P2,P1,pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{115}event PolicyAccepted(P2,P1,pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{116}event PolicyAcceptedKeys(P2,P1,pk(sk2_2),pk(sk1_1),pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{117}event PoliciesComplete(P2)

{120}event FullJoin(P2,mkstar(OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{121}event ConfirmationSent(P2,P1,hstar(mkstar(OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))))

(P2,P1,~M_14)

(P2,P1,~M_14)

{39}event ConfirmationAuthenticated(P1,P2,hstar(mkstar(OnlyGood,OnlyGood,Req(exact(mE1),exacts1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{41}event ConfirmationAccepted(P1,P2,hstar(mkstar(OnlyGood,OnlyGood,Req(exact(mE1),exacts1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{42}event Confirm(P1,mkstar(OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

Beginning of process attest

~M_4 = pk(skE_2)

{48}new challenge_4

{49}event ChallengeSent(P1,pk(skE_2),challenge_4)

~M_15 = challenge_4

~M_15 = challenge_4

{183}event QuoteProduced(pk(skE_2),challenge_4, Identity(mBad,s1,prodA,V2),Good)

~M_16 = sign(teeQuote(Identity(mBad,s1,prodA,V2), Good,hchal(challenge_4,pk(skE_2))),akTEE_1)

~M_16 = sign(teeQuote(Identity(mBad,s1,prodA,V2), Good,hchal(challenge_4,pk(skE_2))),akTEE_1)

{52}event QuoteReceived(P1,E1,pk(skE_2),challenge_4)

{54}event QuoteSignatureChecked(P1,E1,pk(skE_2), challenge_4,Identity(mBad,s1,prodA,V2),Good)

{56}event QuoteVerified(P1,E1,pk(skE_2),challenge_4, Identity(mBad,s1,prodA,V2),Good)